

Bayesian ML: project topics

Rémi Bardenet, Julyan Arbel, Gabriel Victorino Cardoso

February 5, 2025

1 Nature of the project

A group is made of two students. Each group is to pick one of the papers listed below, and each paper can be chosen only once. The assignment is on a first-come first-served basis. It is possible to get the maximum grade with a paper identified as “short/easy”, but there will be a small bonus for picking a paper identified as “long/difficult”: use this option to decide how much time to spend on your BML project compared to other courses. You can also choose to work on a paper of your choice, subject to our explicit approval; in that case, we trust you not to pick a paper that you are or have been working on in another class.

The whole point is to read your paper with a critical mind. For the paper your group will have chosen, you should: (1) explain the contents of the paper at a high level, (2) emphasize the strong and weak points of the paper, and (3) apply it to real data of your choice when applicable. We also ask you to (4) be creative and add something insightful that is not in the original paper: this can be a theoretical point, an illustrative experiment, etc. State your creative contribution explicitly in the introduction of your report.

2 Assignment of papers

As a first step, we ask each group to fill the spreadsheet at

<https://lite.framacalc.org/euwu5cehat-a5rr>

with the title of the paper, a link to it (if available), and the composition of the group. Please fill the form **before 23:59 on Wednesday 12 February**.

3 Format of the deliverable

Please have each group send

- one report as a pdf (≤ 5 pages, everything included) in the [NeurIPS template](#),
- the link to a [GitHub](#) or [GitLab](#) repository containing your code and a detailed readme file with instructions to (compile/install and) run the code.

to [all three teachers](#)¹ **no later than 20 March, 23:59**. We will then meet on 27 March for presentations (format TBD depending on number of groups).

4 Proposed papers

Short and “easy” papers

- [Easy1] Carlos M Carvalho, Nicholas G Polson, and James G Scott. Handling sparsity via the horseshoe. In *Artificial Intelligence and Statistics*, pages 73–80. PMLR, 2009.
- [Easy2] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux - effortless Bayesian deep learning. In *Advances in Neural Information Processing Systems*, 2021.
- [Easy3] Gintare Karolina Dziugaite and Daniel M Roy. Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data. *arXiv preprint arXiv:1703.11008*, 2017.
- [Easy4] Tomas Geffner, George Papamakarios, and Andriy Mnih. Compositional score modeling for simulation-based inference. In *International Conference on Machine Learning*, pages 11098–11116. PMLR, 2023.
- [Easy5] Alexandros Graikos, Nikolay Malkin, Nebojsa Jojic, and Dimitris Samaras. Diffusion models as plug-and-play priors. *Advances in Neural Information Processing Systems*, 35:14715–14728, 2022.
- [Easy6] Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. *Uncertainty in Artificial Intelligence (UAI)*, 2018.
- [Easy7] Pavel Izmailov, Sharad Vikram, Matthew D Hoffman, and Andrew Gordon Gordon Wilson. What are Bayesian neural network posteriors really like? In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4629–4640. PMLR, 18–24 Jul 2021.
- [Easy8] Ghassen Jerfel, Serena Wang, Clara Wong-Fannjiang, Katherine A Heller, Yian Ma, and Michael I Jordan. Variational refinement for importance sampling using the forward Kullback-Leibler divergence. In *Uncertainty in Artificial Intelligence*, pages 1819–1829. PMLR, 2021.
- [Easy9] Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being Bayesian, even just a bit, fixes overconfidence in ReLU networks. In *International Conference on Machine Learning*, pages 5436–5446, 2020.
- [Easy10] Sanae Lotfi, Pavel Izmailov, Gregory Benton, Micah Goldblum, and Andrew Gordon Wilson. Bayesian model selection, the marginal likelihood, and generalization.

¹If the link does not work, this means remi.bardenet@gmail.com, gabriel.victorino.cardoso@minesparis.psl.eu and julyan.arbel@inria.fr.

In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*. PMLR, 2022.

- [Easy11] Sokhna Diarra Mbacke, Florence Clerc, and Pascal Germain. Statistical guarantees for variational autoencoders using PAC-Bayesian theory. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 56903–56915. Curran Associates, Inc., 2023.
- [Easy12] Tim G. J. Rudner, Sanyam Kapoor, Shikai Qiu, and Andrew Gordon Wilson. Function-Space Regularization in Neural Networks: A Probabilistic Perspective. In *International Conference on Machine Learning*, 2023.
- [Easy13] Jiashuo Wang, Haozhao Wang, Shichao Sun, and Wenjie Li. Aligning language models with Human preferences via a Bayesian approach. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 49113–49132. Curran Associates, Inc., 2023.

Long or difficult papers

- [Hard1] S. Agapiou, O. Papaspiliopoulos, D. Sanz-Alonso, and A. M. Stuart. Importance Sampling: Intrinsic Dimension and Computational Cost. *Statistical Science*, 32(3):405–431, 2017.
- [Hard2] Filippo Ascolani, Antonio Lijoi, Giovanni Rebaudo, and Giacomo Zanella. Clustering consistency with Dirichlet process mixtures. *Biometrika*, 2022.
- [Hard3] A. Belhadji, R. Bardenet, and P. Chainais. Kernel interpolation with continuous volume sampling. In *International Conference on Machine Learning (ICML)*, 2020.
- [Hard4] Anirban Bhattacharya, Debdeep Pati, and Yun Yang. On the convergence of coordinate ascent variational inference. *The Annals of Statistics*, 2025.
- [Hard5] Trevor Campbell and Tamara Broderick. Automated scalable Bayesian inference via Hilbert coresets. *Journal of Machine Learning Research*, 20(15):1–38, 2019.
- [Hard6] Ismaël Castillo and Paul Egels. Posterior and variational inference for deep neural networks with heavy-tailed weights. *arXiv preprint arXiv:2406.03369*, 2024.
- [Hard7] Randal Douc, Pierre E Jacob, Anthony Lee, and Dootika Vats. Solving the Poisson equation using coupled markov chains. *arXiv preprint arXiv:2206.05691*, 2022.
- [Hard8] David T. Frazier, Ryan Kelly, Christopher Drovandi, and David J. Warne. The statistical accuracy of neural posterior and likelihood estimation. *arXiv preprint arXiv:2411.12068*, 2024.
- [Hard9] P. Grünwald and T. Van Ommen. Inconsistency of Bayesian inference for misspecified linear models, and a proposal for repairing it. *Bayesian Analysis*, 12(4):1069–1103, 2017.

- [Hard10] Yuqi Gu and David B Dunson. Bayesian pyramids: identifiable multilayer discrete latent structure models for discrete data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(2):399–426, 2023.
- [Hard11] Matthew D Hoffman and Andrew Gelman. The no-U-turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(1):1593–1623, 2014.
- [Hard12] Pierre E Jacob, John O’Leary, and Yves F Atchadé. Unbiased Markov chain Monte Carlo methods with couplings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(3):543–600, 2020.
- [Hard13] Mohammad Emtiyaz Khan and Havard Rue. The Bayesian learning rule. *Journal of Machine Learning Research*, 24(281):1–46, 2023.
- [Hard14] Jeremias Knoblauch, Jack Jewson, and Theodoros Damoulas. An optimization-centric view on Bayes’ rule: Reviewing and generalizing variational inference. *The Journal of Machine Learning Research*, 23(1):5789–5897, 2022.
- [Hard15] A. Korba, A. Salim, M. Arbel, G. Luise, and A. Gretton. A non-asymptotic analysis for Stein variational gradient descent. *Advances in Neural Information Processing Systems*, 33:4672–4682, 2020.
- [Hard16] Jeffrey W Miller and Matthew T Harrison. Inconsistency of Pitman-Yor process mixtures for the number of components. *The Journal of Machine Learning Research*, 15(1):3333–3370, 2014.
- [Hard17] T. Papamarkou, J. Hinkle, M. T. Young, and D. Womble. Challenges in Markov chain Monte Carlo for Bayesian neural networks. *Statistical Science*, 37(3):425–442, 2022.
- [Hard18] Marcelo Pereyra. Proximal Markov chain Monte Carlo algorithms. *Statistics and Computing*, 26:745–760, 2016.
- [Hard19] Kolyan Ray and Botond Szabó. Variational Bayes for high-dimensional linear regression with sparse priors. *Journal of the American Statistical Association*, 117(539):1270–1281, 2022.
- [Hard20] V Spokoiny and M Panov. Accuracy of Gaussian approximation for highdimensional posterior distributions. *Bernoulli*, 2025.
- [Hard21] Rong Tang and Yun Yang. On empirical Bayes variational autoencoder: An excess risk bound. In Mikhail Belkin and Samory Kpotufe, editors, *Proceedings of Thirty Fourth Conference on Learning Theory*, volume 134 of *Proceedings of Machine Learning Research*, pages 4068–4125. PMLR, 15–19 Aug 2021.
- [Hard22] Achille Thin, Nikita Kotelevskii, Arnaud Doucet, Alain Durmus, Eric Moulines, and Maxim Panov. Monte Carlo variational auto-encoders. In *International Conference on Machine Learning*, pages 10247–10257. PMLR, 2021.